

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

METR (Model Evaluation & Threat Research) | March 2025

A New Metric: 50% Task-Completion Time Horizon

1. Create diverse task suite (170 tasks)

-
- 2. Measure human timing & AI success rates
-
- 3. Fit logistic model to find 50% success duration
-

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Task Suite Composition:

- HCAST: 97 tasks
- SWAA: 66 tasks
- RE-Bench: 7 tasks

Task Family	Human Time	Description
find_shell_script	3 seconds	Which file is a shell script?
wikipedia_research	1 minute	Research simple factual info
oxdna_simple	9 minutes	Detect/fix bug in MD simulation
munge_data	56 minutes	Python script to transform JSON
cuda_backtesting	8 hours	Custom CUDA kernels for 30x speedup

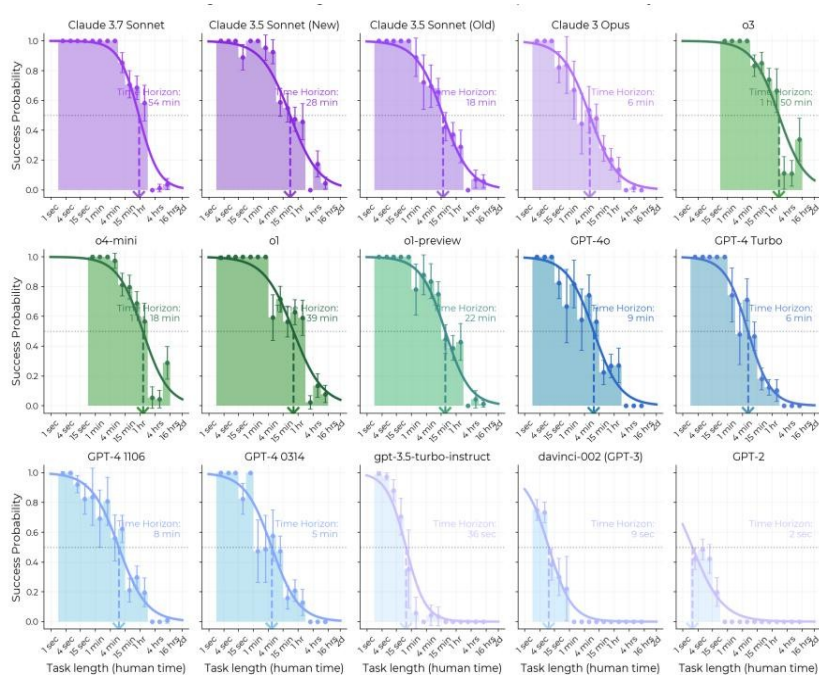
AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years

Key Metrics:

- Doubling time: ~207 days (~7 months)
- $R^2 = 0.97$ (highly consistent trend)

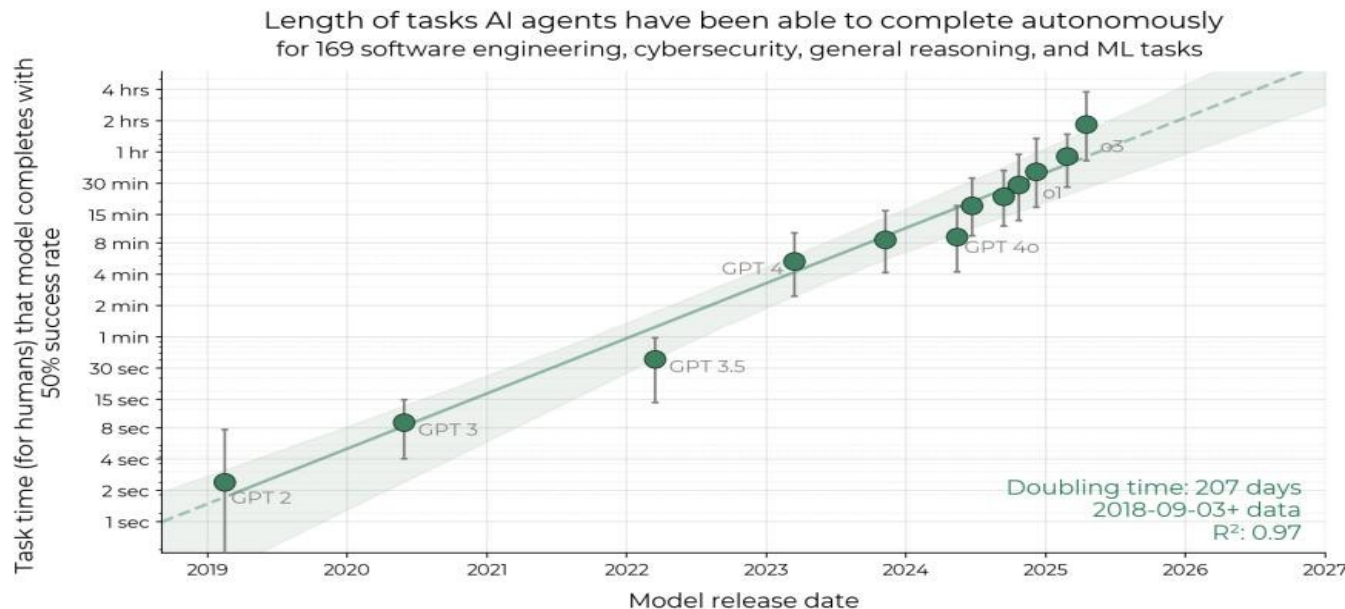
Trajectory:

- GPT-2 (2019): ~2 seconds
- GPT-4 (2023): ~5 minutes
- o3 (2025): ~1 hour 50 minutes



Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models like Claude 3.7 Sonnet (54m) and o3 (1h50m) push the curve significantly rightward compared to older models.



Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Key Drivers of Progress

- Improved logical reasoning
- Better tool use
- Greater reliability in execution
- Self-awareness
- Ability to adapt to mistakes

Current Limitations

- Poor performance on unstructured, 'messier' tasks
- External validity: Tasks may not perfectly represent average real-world intellectual labor

Projections & Caveats: Extrapolation Predicts Month-Long Task Automation by 2028–2031

- >1 Month Horizon (167 hours): Expected between mid-2028 and mid-2031
-
- • Acceleration: The 2023–2025 growth rate is ~20% faster than the overall 2019–2025 trend
-
- • Real-World Impact: Suggests automation of entire month-long software projects in the near future
-
- • Caveat: Assuming external validity holds and trends don't plateau

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

- 1. Robust Metric:** 50% time horizon provides intuitive tracking ($R^2=0.97$).
- 2. Accelerating Trend:** Doubling every ~7 months, possibly faster since 2024.
- 3. Current Frontier:** Models like o3 can complete ~2 hours of human-equivalent work.
- 4. External Validity:** Still an open question on generalizability to all jobs.
- 5. Next Steps:** Unstructured tasks remain a hurdle, but exponential growth demands proactive safety governance.